5.9.3 Common atmospheric pollutants and their sources

5.9.3.1 Atmospheric pollutants from fuels

Content	Key opportunities for skills development
The combustion of fuels is a major source of atmospheric pollutants. Most fuels, including coal, contain carbon and/or hydrogen and may also contain some sulfur. The gases released into the atmosphere when a fuel is burned may include carbon dioxide, water vapour, carbon monoxide, sulfur dioxide and oxides of nitrogen. Solid particles and unburned hydrocarbons may also be released that form particulates in the atmosphere. Students should be able to: • describe how carbon monoxide, soot (carbon particles), sulfur dioxide and oxides of nitrogen are produced by burning fuels	
• predict the products of combustion of a fuel given appropriate information about the composition of the fuel and the conditions in which it is used.	WS 1.2

5.9.3.2 Properties and effects of atmospheric pollutants

Content	Key opportunities for skills development
Carbon monoxide is a toxic gas. It is colourless and odourless and so is not easily detected. Sulfur dioxide and oxides of nitrogen cause respiratory problems in humans and cause acid rain. Particulates cause global dimming and health problems for humans.	
Students should be able to describe and explain the problems caused by increased amounts of these pollutants in the air.	WS 1.4